

**CORK INSTITUTE OF TECHNOLOGY**  
**INSTITIÚID TEICNEOLAÍOCHTA CHORCAÍ**

**Semester Project 2021/22**

**Module Title: Times Series & FA**

**Module Code:** STAT9005

**School:** Mathematical Science and Informatics

**Programme Title:** MSc in Data Science and Analytics

**Internal Examiner(s):** Francisco Hernandez

**Instructions:**

This assessment is worth **40%** of your final grade in STAT9005.

**(30%)** Report on case studies (**20% for Time Series** and 10% for FA).

**(10%)** Recorded Presentation.

**Deadline:** Submit via Canvas the report in PDF format, the RStudio script and a ten minutes presentation of the work, no later than **Sunday 22<sup>nd</sup> of May 2022** at 22 hours. The final file submitted should include the report, code and presentation for both parts (Time Series and Dimensionality Reduction).

You should also include a declaration as part of your submission that all work contained in the submission was your own except for others work which is clearly referenced.

# **Principal Component Analysis & Factor Analysis**

A dataset is provided in the file “FA\_STAT9005\_Project.sav” in Canvas.

## **Background Information**

Cars were selected at random from among 93 American passenger car models that were listed in both the *Consumer Reports* issue and the *PACE Buying Guide* magazines. Pickup trucks and Sport/Utility vehicles were eliminated due to incomplete information and duplicate models were listed at most once.

The variables relevant to this project are defined as follows:

| <b>Variable Name</b>      | <b>Description</b>                                                           |
|---------------------------|------------------------------------------------------------------------------|
| <i>Origin</i>             | <i>Manufacturing company origin. 1="non-USA" or 2="USA"</i>                  |
| <i>Price</i>              | <i>Price (in \$1,000)</i>                                                    |
| <i>Type</i>               | <i>Level: 1="Small" 2="Sporty" 3="Compact" 4="Midsize" 5="Large" 6="Van"</i> |
| <i>Engine Size</i>        | <i>Engine size (litres).</i>                                                 |
| <i>RPM</i>                | <i>RPM (revs per minute at maximum horsepower)</i>                           |
| <i>Fuel.Tank.Capacity</i> | <i>Fuel tank capacity (US gallons)</i>                                       |
| <i>MPG.city</i>           | <i>City MPG (miles per US gallon by EPA rating)</i>                          |
| <i>Weight</i>             | <i>Weight (pounds)</i>                                                       |

## **Preliminary analysis**

Open the SPSS file into R or RStudio and provide a preliminary description of the dataset using a descriptive summary.

Explain the commands used in R and the manipulation performed to prepare the data for a data reduction analysis.

Study the correlation between the variables of the dataset using the correlation matrix and the command “*pairs*”. Plot the relationship of 3 pairs of variables of your choice and fit a regression line to them.

Generate 3D plots to represent the correlation between the following variables. Combine the variables in an appropriate manner to plot a 2D-regression plane on each of the 3D plots for the following regressions:

*MPG.city on Cylinders and EngineSize*

*Horsepower on Cylinders and EngineSize*

Based on the relationship between the variables, what would be (if any) your initials guess regarding the latent variables underlying in the dataset?

Carry out an appropriate test to check whether the dataset is suitable for implementing data reduction techniques.

Comment on your results.

## **Principal Components**

Perform PCA analysis to the dataset using base R commands and another using any R library of your choice. For the model, identify and compare the results of:

- a. Eigenvalues and eigenvectors for each component.
- b. Amount of information explained by each component as well as the cumulative variance.
- c. Identify the resulting factor loadings and scores in the analysis. Explain the way these elements are computed in PCA.
- d. Study the correlation between the original variables and each principal component. What variables are represented by the first two principal components? Use factor matrix and biplot to comment and support your findings.
- e. Use a screeplot to decide how many components to retain. Explain why you are choosing these components. Is this decision supported by what have you observed in the previous question using the factor loading and biplots?
- f. Use the PC scores to interpret the relation between the observations and each of the components. Link the values of the scores on the matrix with the coordinates of the observations (use the biplot for coordinates to observe the position in relation with the PC's)
- g. Use the libraries *"FactoMineR"* and *"factoextra"*, and the commands *"fviz\_eig"*, *"fviz\_pca\_var"*, *"fviz\_pca\_ind"* from the libraries to represent:
  - A screeplot and correlation circle of the dataset.
  - A biplot representing the cars by "Origin" – USA and NonUSA.

Do you observe any difference in the biplot between the USA manufacturers and the rest of carmakers? Which variables do you think are involved?

## **Factor Analysis**

Carry out a Factor analysis on the dataset using the “factanal” from the library “psych”. Perform the analysis considering 1, 2 and 3 underlying variables and comment on the results.

Use the library “nFactors” to determine the number of factors to extract. Is this consistent with what you obtained from previous PCA steps?

Interpret and comment on the factor loadings you have obtained.

Use “fa” command from library “FactoMineR” and compare the result with your previous one. Use different methods for factor extraction (e.g. Maximum likelihood and Principal axis) and compare the results. Do you find any difference in the factors obtained using different methods?

Use the command “fa.diagram” to obtain a path diagram of the items’ factor loadings. Build two diagrams using different number of factors. Interpret and compare the results for the different number of factors chosen.

## **Models**

Write the final PCA model for this dataset.

### **Second analysis**

Split the dataset into two groups by its origin (USA-cars and NonUSA-cars). (Hint: use the “split” command in R)

Repeat the PCA and FA analyses on one of these two datasets (USA or Non-USA cars).

Do you find any differences with the results obtained previously for the full dataset?